

 Learn  Practice

Chemical Bonding & Hybridisation — D4

Issue #38 Candidate Specimen: Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In formamide, $\text{H}-\text{C}(=\text{O})-\text{NH}_2$, a local Lewis count gives nitrogen three sigma bonds and a lone pair. You are also given that the approximately planar amide arrangement permits nitrogen-lone-pair donation into the adjacent carbonyl pi system, which requires a suitably aligned p-like orbital. Choose a model that accounts for both the sigma framework and delocalisation. Resolve the

Representation

Fixed connectivity: $\text{H}-\text{C}(=\text{C})$
End-on view along C-N; cc

Connectivity

apparent conflict with an automatic 'four domains therefore sp³' label and state the limits of your chosen hybridisation description.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Common wrong route

Four Lewis entries overrule supplied delocalisation conditions.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Resolve conflict between a local Lewis inventory and supplied aligned delocalisation.

H1 Representation

Guided

A hybrid label describes a chosen basis; it does not independently cause donation.

H2 First move

Guided

Account for sigma framework, donor orientation and adjacent pi acceptor before qualifying the label.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Give one observation that constrains the model and one claim it cannot uniquely prove.

Need the concept again?

Delocalisation-sensitive
hybridisation boundaries

Answer and working

Hide

Reasoning route

1

Understand

Record the Lewis inventory and the supplied donation condition separately.

Uses

In formamide, $\text{H}-\text{C}(=\text{O})-\text{NH}_2$, a local Lewis count gives nitrogen three sigma bonds and a lone pair. You are also given that the approximately planar amide arrangement permits nitrogen-lone-pair donation into the adjacent carbonyl pi system, which requires a suitably aligned p-like orbital. Choose a model that accounts for both the sigma framework and delocalisation. Resolve the apparent conflict with an automatic 'four domains therefore sp^3 ' label and state the limits of your chosen hybridisation description.

Why valid: N has three bonds plus a lone pair, but the task also supplies approximately planar donation into an adjacent carbonyl pi system.

Result: A four-entry Lewis ledger does not settle the electronic model.

2 Connect

Key move

Choose a basis that represents both jobs.

Uses A four-entry Lewis ledger does not settle the electronic model.

Why valid: Three in-plane sp²-like sigma directions can coexist with an aligned p-like lone-pair donor under the supplied model.

Result: A planar sigma framework plus a p-like donor connected to the carbonyl pi space.

3 Calculate

Explain why automatic sp³ assignment is inadequate.

Uses

A planar sigma framework plus a p-like donor connected to the carbonyl pi space.

Why valid: A localized tetrahedral rule omits the stated delocalisation and alignment conditions.

Result: The local count is a useful ledger, not an overriding proof of the basis.

4 Interpret

Bound and counterfactually test the description.

Uses The local count is a useful ledger, not an overriding proof of the basis.

Why valid: Twisting or interrupting the donor-acceptor path changes the applicability; geometry alone does not uniquely measure hybrid coefficients.

Result: An introductory sp²-like account is conditional, not unique observed orbital reality.

Approximately planar amide N can use an introductory sp²-like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.

Independent check

Remove the hybridisation label: the alignment-plus-acceptor explanation must still describe the supplied donation condition.

Test the rule on a changed case

Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?

Your prediction and reason

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Conformational torsion and p-orbital alignment

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For the formamide situation supplied in Q1, construct one qualitative representation connecting the Lewis lone pair on nitrogen, the nitrogen orbital used for donation and the carbonyl pi system. Then construct the corresponding approximately 90-degree-

Representation

Fixed connectivity: H-C(=C

End-on view along C-N; ce

Connectivity

twisted arrangement about the C–N bond. Preserve atom connectivity and distinguish changes in orbital alignment from changes in formal electron count. Do not claim these sketches alone measure the donation strength.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Common wrong route

Torsion changes the formal electron inventory.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Construct both arrangements while preserving connectivity and electron inventories.

H1 Representation

Guided

Torsion changes relative orientation, not the number of atoms or formal electrons.

H2 First move

Guided

Draw a fixed carbonyl reference and an N donor direction rotated about the declared C-N axis.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Count valence and total electrons for CH₃NO before and after torsion. Identify which geometric relation changes.

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Sketch a fixed-connectivity 45-degree comparison without claiming measured donation.

Need the concept again?

Conformational torsion and p-orbital alignment

Answer and working

Hide

Reasoning route

1

Understand

Inventory formamide electrons and connectivity.

Uses

For the formamide situation supplied in Q1, construct one qualitative representation connecting the Lewis lone pair on nitrogen, the nitrogen orbital used for donation and the carbonyl pi system. Then construct the corresponding approximately 90-degree-twisted arrangement about the C-N bond. Preserve atom connectivity and

distinguish changes in orbital alignment from changes in formal electron count. Do not claim these sketches alone measure the donation strength.

Why valid: CH₃NO supplies 4+3+5+6 valence electrons and 6+3+7+8 total electrons.

Result: 18 valence; 24 total; H-C(=O)-NH₂ connectivity fixed.

2 Connect

Key move

Construct the supplied aligned donor/carbonyl pi account.

Uses 18 valence; 24 total; H-C(=O)-NH₂ connectivity fixed.

Why valid: The localized donor pair plus carbonyl pi pair gives four electrons in a three-centre qualitative pi space.

Result: Aligned N p-like donor and C/O p directions; four model pi-space electrons.

3 Calculate

Rotate the N donor direction about C-N toward 90 degrees.

Uses Aligned N p-like donor and C/O p directions; four model pi-space electrons.

Why valid: Relative torsion changes orientation against the fixed carbonyl reference; it does not change element counts.

Result: Approximately orthogonal donor/acceptor directions with unchanged connectivity and electron ledger.

4 Interpret

Check what the two sketches can establish.

Uses

Approximately orthogonal donor/acceptor directions with unchanged connectivity and electron ledger.

Why valid: Orientation distinguishes alignment, while donation strength needs an experiment or specified calculation.

Result: Both drawings retain 18 valence/24 total electrons; no measured donation strength is claimed.

Formamide CH₃NO has 18 valence electrons and 24 total electrons. The supplied three-centre donor/carbonyl pi model has four electrons. A 90-

degree twist changes alignment, not these inventories.

Independent check

Independent element sums: valence $4+3+5+6=18$; total $6+3+7+8=24$. Both sketches must preserve them.

Test the rule on a changed case

Count valence and total electrons for CH_3NO before and after torsion. Identify which geometric relation changes.

Your prediction and reason

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Conformational torsion and p-orbital alignment

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies a qualitative model result: with atom connectivity fixed, twisting a planar amide arrangement toward 90 degrees reduces the nitrogen-to-carbonyl orbital alignment

Representation

Fixed connectivity: $\text{H}-\text{C}(=\text{C})$
End-on view along C-N; cc

Connectivity

and the delocalisation contribution. Explain that result through orbital overlap. Assess whether 'the hybridisation label changes, so donation disappears' is a sufficient explanation, and identify which parts of your account are model assumptions rather than measured facts.

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

—

Common wrong route

A changed hybrid label is a causal explanation of donor strength.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Explain the supplied trend without replacing a mechanism with a label.

H1 Representation

Guided

Relative donor/acceptor alignment is a model variable; donation magnitude needs additional physics.

H2 First move

Guided

Connect changed geometry to overlap, then tag each statement as given, model or observation.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: If two sketches have the same label but different donor alignment, must their donation be identical? If the label changes alone, has a mechanism been supplied?

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: State which additional model would be needed for a numerical donation or barrier estimate.

Need the concept again?

Conformational torsion and p-orbital alignment

Answer and working

Hide

Reasoning route**1****Understand**

Use the supplied model result as a conditional premise.

Uses

The benchmark supplies a qualitative model result: with atom connectivity fixed, twisting a planar amide arrangement toward 90 degrees reduces the nitrogen-to-carbonyl orbital alignment and the delocalisation contribution. Explain that result through orbital overlap. Assess whether 'the hybridisation label changes, so donation disappears' is a sufficient explanation, and identify which parts of your account are model assumptions rather than measured facts.

Why valid: The benchmark specifies reduced alignment and delocalisation contribution toward 90-degree twist.

Result: Connectivity fixed; relevant orbital directions rotate relative to each other.

2 Connect

Key move

Connect torsion to the overlap mechanism.

Uses Connectivity fixed; relevant orbital directions rotate relative to each other.

Why valid: Parallel p-like donor/acceptor directions support lateral interaction; orthogonal directions reduce that particular alignment.

Result: Reduced alignment explains the stated qualitative contribution change.

3 Calculate

Test the hybrid-label-only explanation.

Uses Reduced alignment explains the stated qualitative contribution change.

Why valid: A label does not encode the adjacent acceptor or their relative orientation.

Result: Renaming hybridisation alone is insufficient; the alignment pathway supplies the mechanism.

4 Interpret

Separate model contribution from measured total energy.

Uses

Renaming hybridisation alone is insufficient; the alignment pathway supplies the mechanism.

Why valid: Other energy terms and relaxation can also vary under torsion.

Result: No unique physical basis, numerical barrier or full total-energy law follows from this qualitative account.

Twisting reduces relevant alignment and the supplied delocalisation contribution. A changing hybrid label alone explains neither mechanism nor a quantitative barrier; total torsional energy has additional contributions.

Independent check

Counterfactual: remove the word hybridisation while retaining changed orientation; the mechanistic statement survives.

Test the rule on a changed case

If two sketches have the same label but different donor alignment, must their donation be identical? If the label changes alone, has a mechanism been supplied?

Your prediction and reason

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Local domain counting versus conjugated networks

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

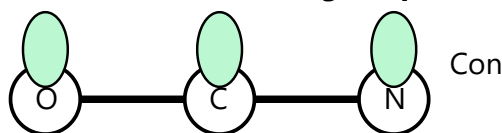
Provenance: Owner-supplied question

ATTEMPT FIRST

Compare two nitrogen sites: A is the amide nitrogen in $\text{H}-\text{C}(=\text{O})-\text{NH}_2$; B is the amine nitrogen in $\text{H}-\text{C}(=\text{O})-\text{CH}_2-$

Representation

Site A · Direct amide: continuous aligned p-orbitals



NH2. In the simplified introductory model, the intervening saturated CH₂ group has no continuous aligned p-orbital pathway. Develop a decision procedure combining local sigma/lone-pair counting, adjacency, orbital alignment and the model's validity conditions. Apply it to A and B, then state a counterfactual structural change that would require you to reassess your procedure. Do not equate spatial proximity with conjugation.

Site A: Continuous pi highway

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Common wrong route

Spatial proximity equals an orbital-connected p pathway.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0

Key concept

Guided

Build a decision procedure that can fail its own applicability test.

H1 Representation Guided

Local count and a continuous aligned pathway are independent checks.

H2 First move Guided

Test locality, adjacency, alignment and stated model conditions; invent a structural counterfactual yourself.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Would folding the saturated-spacer molecule closer create the same continuous p pathway as removing the spacer?

H4 Assembly checkpoint Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Specify a counterfactual bridge and test all procedure conditions again.

Need the concept again?

Local domain counting versus conjugated networks

Answer and working

Hide

Reasoning route

1 Understand

Apply the same N Lewis ledger at both sites.

Uses

Compare two nitrogen sites: A is the amide nitrogen in $\text{H}-\text{C}(=\text{O})-\text{NH}_2$; B is the amine nitrogen in $\text{H}-\text{C}(=\text{O})-\text{CH}_2-\text{NH}_2$. In the simplified introductory model, the intervening saturated CH_2 group has no continuous aligned p-orbital pathway. Develop a decision procedure combining local sigma/lone-pair counting, adjacency, orbital alignment and the model's validity conditions. Apply it to A and B, then state a counterfactual structural change that would require you to reassess your procedure. Do not equate spatial proximity with conjugation.

Why valid: Both neutral N sites have three sigma bonds and one lone pair.

Result: Local count alone does not distinguish A from B.

2 Connect

Key move

Trace donor-to-carbonyl adjacency and p availability.

Uses Local count alone does not distinguish A from B.

Why valid: A has direct $\text{N}-\text{C}(=\text{O})$ adjacency; B contains the supplied saturated CH_2 interruption.

Result: A supports the stated contiguous donor/acceptor path; B lacks that ordinary path.

3 Calculate

Test alignment and validity before applying a label.

Uses

A supports the stated contiguous donor/acceptor path; B lacks that ordinary path.

Why valid: Donation requires suitable orbital orientation as well as adjacency; spatial proximity does not create conjugation.

Result: Use conditional delocalised accounting at A and localized amine accounting at B within the given model.

4 Interpret

Construct a structural counterfactual.

Uses

Use conditional delocalised accounting at A and localized amine accounting at B within the given model.

Why valid: Replacing the saturated interruption with a p-bearing aligned bridge changes the path test.

Result: Reassess the model; absence of this ordinary path is not an absolute zero-interaction theorem.

Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and absence of this path is not an absolute zero-interaction theorem.

Independent check

Changed case: replace a saturated interruption by a suitably aligned unsaturated bridge; reassess connectivity and orbital availability rather than reusing the answer.

Test the rule on a changed case

Would folding the saturated-spacer molecule closer create the same continuous p pathway as removing the spacer?

Your prediction and reason

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student claims:

'Hybridisation uniquely determines bond angles and any deviation proves the hybridisation label wrong.' The benchmark supplies approximate observations: CH₄ tetrahedral about 109.5 degrees; NH₃ pyramidal about 107 degrees; H₂O bent about 104.5 degrees; all three commonly use an introductory sp³ description at the central atom. It also supplies approximately planar amide nitrogen with lone-pair delocalisation. Construct and test a qualified replacement claim using these observations, local domain counting, lone-pair effects and the amide boundary case. State what further evidence would be needed before treating a qualitative orbital picture as a unique physical account.

Representation

Fixed connectivity: H-C(=C

End-on view along C-N; ce

Connectivity

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model,

Common wrong route · open if you want a warning

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not an autonomous physical observable.

Common wrong route

A hybrid label uniquely determines every angle and physical orbital basis.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept Guided

Test uniqueness, rather than merely finding one familiar angle.

H1 Representation Guided

Several shapes/angles can share an introductory four-domain classification.

H2 First move Guided

Use supplied counterexamples and the amide boundary to qualify both predictive scope and evidence.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Can one four-domain label predict the same nuclear shape for CH₄ and H₂O? Does observing a shape uniquely establish an orbital basis?

H4 Assembly checkpoint Guided

Reserve the last judgement for yourself. Try this boundary/transfer

task: Name evidence needed before claiming a unique physical electronic description.

Need the concept again?

Delocalisation-sensitive hybridisation boundaries

Answer and working

Hide

Reasoning route

1

Understand

Compare the supplied angles with their shared introductory label.

Uses

A student claims: 'Hybridisation uniquely determines bond angles and any deviation proves the hybridisation label wrong.' The benchmark supplies approximate observations: CH₄ tetrahedral about 109.5 degrees; NH₃ pyramidal about 107 degrees; H₂O bent about 104.5 degrees; all three commonly use an introductory sp³ description at the central atom. It also supplies approximately planar amide nitrogen with lone-pair delocalisation. Construct and test a qualified replacement claim using these observations, local domain counting, lone-pair effects and the amide boundary case. State what further evidence would be needed before treating a qualitative orbital picture as a unique physical account.

Why valid: CH₄, NH₃ and H₂O use sp³-like descriptions but have different lone-pair counts and nuclear shapes.

Result: 109.5, about 107 and about 104.5 degrees are compatible with the supplied shared label.

2

Connect

Key move

Reject the unique-angle implication.

Uses

109.5, about 107 and about 104.5 degrees are compatible with the supplied shared label.

Why valid: A single model label does not fix all interactions that affect geometry.

Result: Angle deviations alone do not falsify every sp³-like description.

3 Calculate

Include the amide boundary in the replacement rule.

Uses Angle deviations alone do not falsify every sp³-like description.

Why valid: An aligned adjacent acceptor and delocalisation can make automatic localized four-domain classification inadequate.

Result: Local counts guide a bounded starting model; lone pairs, alignment and delocalisation require additional checks.

4 Interpret

Specify evidence needed for stronger physical claims.

Uses

Local counts guide a bounded starting model; lone pairs, alignment and delocalisation require additional checks.

Why valid: Nuclear geometry alone does not uniquely recover wavefunctions or a hybrid decomposition.

Result: Use defined spectroscopic/electronic-structure evidence and model assumptions; retain alternative compatible descriptions.

CH₄/NH₃/H₂O share common sp³-like introductory descriptions while their nuclear shapes/angles differ. Geometry constrains models but does not uniquely measure a hybrid decomposition; lone pairs and delocalisation matter.

Independent check

Independent counterexample: a shared four-domain label is compatible with 4,3,2 bonded-atom positions, so it cannot alone uniquely specify atom-only shape.

Test the rule on a changed case

Can one four-domain label predict the same nuclear shape for CH₄ and H₂O? Does observing a shape uniquely establish an orbital basis?

Your prediction and reason

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Local domain counting versus conjugated networks

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

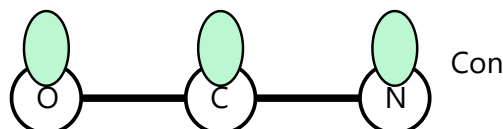
Provenance: Owner-supplied question

ATTEMPT FIRST

Consider the two nitrogen sites A and B in Q4. State the conditions under which the local four-domain counting rule is a useful introductory classification, and the evidence that would make its automatic use inadequate. Distinguish recalling a rule from deciding whether its conditions hold.

Representation

Site A · Direct amide: continuous aligned p-orbitals



Site A: Continuous pi highway

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.

Common wrong route · open if you want a warning

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- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route

Recalling a local rule proves its applicability.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept Guided

Distinguish recalling a rule from establishing its applicability.

H1 Representation Guided

Local counting assumes the localized/domain model is useful for the asked task.

H2 First move Guided

List the conditions and look for supplied delocalisation or unusual occupancy evidence before applying the shortcut.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Both sites have three sigma bonds and a lone pair. Is that alone sufficient to choose identical localized orbital-role descriptions?

H4 Assembly checkpoint Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Separate a recalled count from a model-selection decision in a new conjugated case.

Need the concept again?

Local domain counting versus conjugated networks

Answer and working

Hide

Reasoning route

1 Understand

Recall the local rule with its scope.

Uses

Consider the two nitrogen sites A and B in Q4. State the conditions under which the local four-domain counting rule is a useful introductory classification, and the evidence that would make its automatic use inadequate. Distinguish recalling a rule from deciding whether its conditions hold.

Why valid: Four domains classify ordinary localized introductory centres; recalling it does not show that locality holds.

Result: Starting ledger: three N bonds plus one lone pair.

2 Connect

Key move

Inspect the supplied adjacency condition.

Uses Starting ledger: three N bonds plus one lone pair.

Why valid: A has a carbonyl acceptor directly adjacent; B has a saturated interruption in the ordinary p path.

Result: A needs a delocalisation/alignment test; B supports the local starting model here.

3 Calculate

Identify evidence that defeats automatic application.

Uses

A needs a delocalisation/alignment test; B supports the local starting model here.

Why valid: Suitable donor/acceptor alignment, donation evidence or unusual occupancy add jobs the bond/pair total does not resolve.

Result: Do not infer adequacy from the number four alone.

4 Interpret

Separate known conditions from unknown quantities.

Uses Do not infer adequacy from the number four alone.

Why valid: Neither exact donor strength nor unique hybrid coefficients are supplied by this ledger.

Result: Apply a conditional classification or request the missing electronic evidence.

Four-domain counting is useful for localized introductory classification. Adjacent pi acceptors, aligned donor orbitals, planarity/donation evidence or radical occupancy can require a richer model.

Independent check

Changed check: the same formal lone-pair entry at an amine and an amide can have different orbital-role descriptions under the supplied conditions.

Test the rule on a changed case

Both sites have three sigma bonds and a lone pair. Is that alone sufficient to choose identical localized orbital-role descriptions?

Your prediction and reason

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

Question difficulty: Multi-step reasoning

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

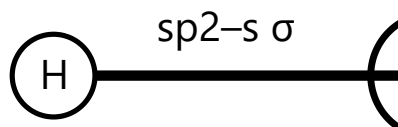
Provenance: Owner-supplied question

ATTEMPT FIRST

A neutral methyl radical, $\text{CH}_3\cdot$, is supplied as approximately planar, with the unpaired electron in a p-like orbital perpendicular to the three C–H sigma directions. Translate that supplied model into an orbital sketch and assess whether 'three sigma bonds always leave an empty p orbital' follows. Keep the given model distinct from a universal claim.

Representation

1 · Planar sp^2 carbon sigma framev



Planar sp^2 framework

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

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Common wrong route

Three sigma bonds always imply an empty residual p orbital.

HINT LADDER • REVEAL ONLY WHAT YOU NEED**Guided support****H0 Key concept**

Guided

Translate occupancy as well as geometry in the supplied radical model.

H1 Representation

Guided

A singly occupied orbital is not an empty orbital or a lone pair.

H2 First move

Guided

Inventory the sigma electrons and remaining electron before drawing the p-like direction.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising:
Compare CH₃ radical (7 valence electrons) with a model CH₃⁺ inventory (6). Must their residual p occupancy be the same?

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Sketch the given radical while explicitly marking the singly occupied orbital.

Need the concept again?

Delocalisation-sensitive
hybridisation boundaries

Answer and working

Hide

Reasoning route

1 Understand

Read the supplied radical geometry and occupancy.

Uses

A neutral methyl radical, $\text{CH}_3\cdot$, is supplied as approximately planar, with the unpaired electron in a p-like orbital perpendicular to the three C–H sigma directions. Translate that supplied model into an orbital sketch and assess whether 'three sigma bonds always leave an empty p orbital' follows. Keep the given model distinct from a universal claim.

Why valid: The benchmark specifies three planar C–H sigma directions and a perpendicular singly occupied p-like orbital.

Result: A three-direction sigma frame plus one unpaired electron.

2 Connect

Key move

Allocate the carbon valence basis.

Uses A three-direction sigma frame plus one unpaired electron.

Why valid: An sp^2 -like sigma allocation uses three functions and leaves one p-like function perpendicular.

Result: Three hybrids plus one residual p-like function; four functions total.

3 Calculate

Place the supplied electron in the residual orbital.

Uses Three hybrids plus one residual p-like function; four functions total.

Why valid: CH_3 radical has $4+3=7$ valence electrons; three sigma pairs use six.

Result: Residual p-like SOMO contains one electron, not zero.

4

Interpret

Compare with a six-electron CH_3^+ model inventory.

Uses Residual p-like SOMO contains one electron, not zero.

Why valid: Removing one electron changes residual occupancy while the three-bond count can remain.

Result: Three sigma bonds alone do not determine whether residual p is empty; the given radical model is not a universal claim.

Supplied planar CH_3 radical has three sigma directions and a perpendicular p-like SOMO with one electron. Three sigma bonds alone do not determine an empty residual p orbital.

Independent check

Independent count: CH_3 radical has $4+3=7$ valence electrons; three sigma pairs use six, leaving one.

Test the rule on a changed case

Compare CH_3 radical (7 valence electrons) with a model CH_3^+ inventory (6). Must their residual p occupancy be the same?

Your prediction and reason

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Two qualitative descriptions of formamide are proposed: a completely localised pyramidal nitrogen lone pair, and an approximately planar nitrogen arrangement with possible lone-pair donation into the carbonyl pi system. Propose observations that could discriminate between them, explain the warranted inferences, and identify at least one inference that geometry alone cannot establish.

Representation

Fixed connectivity: $\text{H}-\text{C}(=\text{C})$
End-on view along C-N; ce

Connectivity

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model, not an autonomous physical observable.

Common wrong route · open if you want a warning

—

Common wrong route

Geometry alone uniquely measures donation and a rotational barrier.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Separate a discriminating observation from the strength of the inference it warrants.

H1 Representation

Guided

A measurement can constrain multiple models without uniquely selecting an orbital decomposition.

H2 First move

Guided

For each proposed observation write prediction, alternatives, conditions and inference limit.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Design a second independent observation and state one alternative explanation.

Need the concept again?

Delocalisation-sensitive
hybridisation boundaries

Answer and working

Hide

Reasoning route

1 Understand

State competing predictions before proposing observations.

Uses

Two qualitative descriptions of formamide are proposed: a completely localised pyramidal nitrogen lone pair, and an approximately planar nitrogen arrangement with possible lone-pair donation into the carbonyl pi system. Propose observations that could discriminate between them, explain the warranted inferences, and identify at least one inference that geometry alone cannot establish.

Why valid: The proposed localized pyramidal and donation-compatible planar accounts differ in nuclear arrangement and possible electronic consequences.

Result: Compare N geometry, relevant bond metrics and condition-specific electronic or dynamic signals.

2 Connect

Key move

Propose controlled geometry and bond measurements.

Uses

Compare N geometry, relevant bond metrics and condition-specific electronic or dynamic signals.

Why valid: Defined species, phase and conditions allow predictions to be compared with uncertainty and alternative causes.

Result: Near-planarity can challenge a strictly pyramidal description; bond metrics add constraints.

3 Calculate

Add an independent electronic or torsional test.

Uses

Near-planarity can challenge a strictly pyramidal description; bond metrics add constraints.

Why valid: A specified spectrum or calculation/experiment can test donation-related or rotation-related predictions under stated assumptions.

Result: A second channel of evidence can discriminate accounts more strongly than geometry alone.

4

Interpret

Limit the warranted inference.

Uses

A second channel of evidence can discriminate accounts more strongly than geometry alone.

Why valid: Compatible geometry can arise under multiple electronic descriptions; strength and barrier require quantitative evidence.

Result: No unique hybrid coefficients, donation magnitude or rotational barrier is measured by a qualitative planar sketch.

Planarity/bond metrics and condition-specific spectroscopic/dynamic evidence can test localized versus delocalised accounts. Geometry alone does not uniquely measure donation, hybrid coefficients or a rotational barrier; exact empirical numbers need species/conditions/source.

Independent check

Use two compatible electronic descriptions of a planar structure as a counterexample to geometry-implies-unique-orbitals.

Test the rule on a changed case

Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

Your prediction and reason

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Conformational torsion and p-orbital alignment

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies a simplified torsional model in which the relevant p-orbital overlap factor varies as $\cos(\theta)$, where θ is the angle between their alignment directions. Compare $\theta = 0, 60$ and 90 degrees, then assess the claim that the molecule's total energy must therefore vary as $\cos(\theta)$. Separate execution of the given model from a justified claim about the physical system.

Representation

Fixed connectivity: H-C(=C)
End-on view along C-N; ce

Connectivity

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model,

Common wrong route · open if you want a warning

—

not an autonomous physical observable.

Common wrong route

The given overlap factor is the total molecular energy.

Need a hint? · guided support

—

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Execute the supplied cosine model and separately test the total-energy inference.

H1 Representation

Guided

An overlap factor is not an energy Hamiltonian or the whole energy budget.

H2 First move

Guided

Compute the three factors, then identify an independent energy term that the input does not constrain.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: If theta stays at 60 degrees while an independent energy term changes, does overlap change? Does total energy necessarily stay fixed?

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: State assumptions needed for a weak-coupling second-order squared-interaction estimate.

Need the concept again?

Conformational torsion and p-orbital alignment

Answer and working

Hide

Reasoning route

1 Understand

Evaluate the supplied overlap factor at the three angles.

Uses

The benchmark supplies a simplified torsional model in which the relevant p-orbital overlap factor varies as $\cos(\theta)$, where θ is the angle between their alignment directions. Compare $\theta = 0, 60$ and 90 degrees, then assess the claim that the molecule's total energy must therefore vary as $\cos(\theta)$. Separate execution of the given model from a justified claim about the physical system.

Why valid: The stated model is $\cos(\theta)$; $\cos(0)=1$, $\cos(60)=1/2$ and $\cos(90)=0$.

Result: Factors: 1, 1/2, 0.

2 Connect

Key move

Identify the missing link from overlap to total energy.

Uses Factors: 1, 1/2, 0.

Why valid: An overlap factor is not a Hamiltonian and supplies no full energy budget.

Result: The proposed total-energy conclusion is not warranted.

3**Calculate**

Construct an explicit counterexample to uniqueness.

Uses The proposed total-energy conclusion is not warranted.

Why valid: At fixed $\theta=60$ an independently specified energy contribution can vary while the overlap factor stays $1/2$.

Result: Same overlap, different total energy; proportionality is not implied.

4**Interpret**

Bound even alternative energy approximations.

Uses Same overlap, different total energy; proportionality is not implied.

Why valid: A squared-interaction estimate needs stated weak-coupling assumptions; a degenerate two-level coupling can instead yield $\epsilon \pm |t_0 \cos(\theta)|$.

Result: No universal cosine or cosine-squared total-energy law; both functions are even, so parity does not select one.

Given factors at 0,60,90 degrees are 1,1/2,0. Both cosine and cosine-squared are even. A degenerate two-orbital Hamiltonian with coupling $t_0 \cos(\theta)$ gives $\epsilon \pm |t_0 \cos(\theta)|$, so no universal cosine-squared total-energy law follows.

Independent check

At $\theta=60$ degrees the factor is fixed at $1/2$ while adding an independently specified energy contribution changes total energy. This falsifies uniqueness without modelling formamide quantitatively.

Test the rule on a changed case

If θ stays at 60 degrees while an independent energy term changes, does overlap change? Does total energy necessarily stay fixed?

Your prediction and reason

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Local domain counting versus conjugated networks

Concept difficulty: Hard concept

Question difficulty: High-transfer / exception-sensitive

Family: Chemical Bonding, Orbital Alignment and Hybridisation Models

Question type: Constructed response

Source: Owner-supplied question

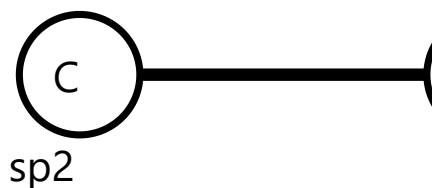
Provenance: Owner-supplied question

ATTEMPT FIRST

A proposed teaching rule says, 'Count sigma bonds and lone pairs, assign a hybridisation label, and the label explains every shape and reactivity difference.' Audit this rule against NH_3 , allene, formamide and the separated amine site $\text{H}-\text{C}(=\text{O})-\text{CH}_2-\text{NH}_2$. Produce a qualified decision framework, identify where local counting is useful, and state what evidence or model is needed beyond it.

Representation

1 · Linear sigma axis and orbital hybridisation



Linear sigma framework

Conditions

- Delocalisation requires continuous, coplanar p-orbital alignment.
- Hybridisation is an idealized localized bonding model,

Common wrong route · open if you want a warning

not an autonomous physical observable.

Common wrong route

A local count and label explain every geometry and reactivity difference.

Need a hint? · guided support

—

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept Guided

Audit a proposed universal rule across cases that stress different boundaries.

H1 Representation Guided

Counts, orbital roles, geometry and physical evidence are distinct layers.

H2 First move Guided

Use NH_3 , allene, amide and separated amine as tests; retain a conditional framework rather than another universal slogan.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Would one bond/lone-pair count plus a label explain NH_3 shape, allene terminal-plane relation and amide donation by the same mechanism?

H4 Assembly checkpoint Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Produce a qualified procedure with an explicit stop-and-look-for-more-evidence branch.

Need the concept again?

Local domain counting versus conjugated networks

Answer and working

Hide

Reasoning route

1 Understand

Audit each case for the explanatory job omitted by the slogan.

Uses

A proposed teaching rule says, 'Count sigma bonds and lone pairs, assign a hybridisation label, and the label explains every shape and reactivity difference.' Audit this rule against NH₃, allene, formamide and the separated amine site H-C(=O)-CH₂-NH₂. Produce a qualified decision framework, identify where local counting is useful, and state what evidence or model is needed beyond it.

Why valid: NH₃ needs lone-pair versus nuclear-shape distinction; allene needs residual-p orientation; formamide needs donation conditions; the separated amine needs a pathway test.

Result: A label alone cannot supply all four explanations.

2 Connect

Key move

Retain local counting as a starting ledger.

Uses A label alone cannot supply all four explanations.

Why valid: Atom-local sigma directions and central pairs give useful introductory information when localized assumptions hold.

Result: Count locally and conserve the valence basis.

3**Calculate**

Add occupancy, adjacency and alignment branches.

Uses Count locally and conserve the valence basis.

Why valid: A residual p function must have a specified occupancy and suitable neighbouring orbital path for the proposed interaction.

Result: Check what orbitals do, rather than treating the hybrid label as their cause.

4**Interpret**

Include a validity and evidence stopping rule.

Uses Check what orbitals do, rather than treating the hybrid label as their cause.

Why valid: Quantitative geometry/reactivity or competing compatible electronic accounts require evidence beyond the ledger.

Result: Use a conditional framework and seek an explicit model or measurement when the simple rule is inadequate.

Use local count as a bounded starting model, then check residual p directions, occupancy, adjacency/alignment and evidence. A label alone explains neither every geometry nor all reactivity. NH₃, allene and amide stress different limitations.

Independent check

Changed check: apply the framework to CH₃ radical; three bonds plus one unpaired electron require an occupancy step absent from the original slogan.

Test the rule on a changed case

Would one bond/lone-pair count plus a label explain NH₃ shape, allene terminal-plane relation and amide donation by the same mechanism?

Your prediction and reason

